Sysslan IT Solutions

Internship Cohort

Data Analysis Task List

About Us

Sysslan IT Solutions

- Sysslan IT Solutions is a leading technology firm recognized for its proficiency in the rapidly evolving domains of cloud computing, blockchain, artificial intelligence (AI), and machine learning (ML).

- The company is dedicated to delivering high-impact projects and customized solutions that address the dynamic requirements of modern businesses.

- Sysslan IT Solutions provides a comprehensive range of products and services, including scalable cloud computing platforms, cutting-edge blockchain solutions, intelligent AI systems, and advanced ML algorithms, designed to empower organizations and drive innovation.

MSME
Micro, Small, & Mediam Enterprises
सूक्ष्म, लघु, एवं मध्यम उद्योग
MSME BUSINESS FORUM

Instructions

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- Update your LinkedIn profile to highlight your accomplishments, such as offer letters or internship certificates from Sysslan IT Solutions. Mention and tag Sysslan IT Solutions in your posts. Use relevant hashtags like #SysslanITSolutions #SysslanExperience #FutureWithSysslan to showcase your professional association.

- Ensure that all work is original and avoid plagiarism or code duplication, as such violations may result in termination of the internship and impact future opportunities with Sysslan IT Solutions.

- Create a video presentation of your completed tasks and share it on LinkedIn. Tag Sysslan IT Solutions and use hashtags like #SysslanInnovation #SysslanITSolutions #SysslanProjects to engage with our professional community.

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SUBMISSION

- Create a professional video highlighting your internship projects and accomplishments.

- Share the video on LinkedIn to demonstrate your work and build credibility with your professional network. Make sure to tag Sysslan IT Solutions in your post so the company is notified of your achievements.

- A submission form will be provided later. In the meantime, continue working on your tasks and maintain separate files for each level of work.

- When posting the video on LinkedIn, use the following hashtags to increase visibility and engagement:
- #SysslanITSolutions #Sysslan #SysslanProjects

- You can also include additional hashtags relevant to your specific internship domain to further showcase your expertise.

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Title: Train Schedule Analysis and Interactive Route Enquiry System Using Python

Project Description:

This project involves performing comprehensive data analysis on train schedules using Python. The goal is to explore, clean, process, and analyze train dataset information to derive meaningful insights about train routes, journey durations, and station traffic. The project gradually progresses from basic data review to advanced analysis and interactive system development, enabling hands-on experience in data handling, visualization, and system implementation.

Key Objectives:

- Understand dataset structure and quality
- Perform data cleaning and preprocessing
- Conduct exploratory and advanced analysis
- Visualize trends and patterns
- Develop an interactive train enquiry system

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Level 1: Basic Data Review

Description: Gain an initial understanding of the dataset, its size, structure, and basic train information.

Tasks & Skills:

- Task 1.1: Overview of dataset – total records & attributes
- Task 1.2: List all trains with their starting and ending stations
- Task 1.3: Calculate the number of stops per train
- Task 1.4: Identify trains with maximum and minimum stops

Skills: Data exploration, Pandas basics, data summarization

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Level 2: Simple Data Processing

Description: Ensure data consistency, compute journey durations, and categorize trains for further analysis.

Tasks & Skills:

- Task 2.1: Standardize schedule fields (arrival & departure times)
- Task 2.2: Compute total journey duration for each train
- Task 2.3: Classify routes as short, medium, or long
- Task 2.4: Generate station-wise train frequency counts

Skills: Data preprocessing, datetime operations, classification, Pandas grouping

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Level 3: Data Quality Checks

Description: Clean and validate the dataset to ensure accurate and reliable analysis.

Tasks & Skills:

- Task 3.1: Handle missing schedule values
- Task 3.2: Remove duplicate train records
- Task 3.3: Verify correct station order in each route
- Task 3.4: Save the verified dataset

Skills: Data cleaning, handling missing values, duplicate removal, data validation

Level 4: Basic Analysis and Visualization

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Description: Analyze journey durations and station traffic and create simple visual representations.

Tasks & Skills:

- Task 4.1: Compare average journey durations across route types
- Task 4.2: Identify high-traffic stations
- Task 4.3: Create basic visualizations for journey duration and station traffic
- Task 4.4: Summarize key observations

Skills: Exploratory data analysis (EDA), Matplotlib, Seaborn, data visualization, insights generation

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Level 5: Advanced Analysis and Visualization

Description: Conduct in-depth analysis using pivot tables and cross-tabulations to uncover detailed trends.

Tasks & Skills:

- Task 5.1: Pivot tables for station-wise train distribution
- Task 5.2: Cross-tabulations to analyze train frequency between stations and routes
- Task 5.3: Visualize pivot and cross-tab results with comparative charts
- Task 5.4: Summarize advanced insights

Skills: Pivot tables, cross-tab analysis, advanced visualization, comparative analytics

Level 6: Final Capstone System

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Description: Build an interactive train enquiry system for real-time route information.

Tasks & Skills:

- Task 6.1: Interactive route-based train enquiry system to display:
  - All available direct trains between a source and destination
  - Estimated journey duration

Skills: Python programming, user input handling, data lookup, application development

How to Contact Us?

To find out more information, please contact us

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